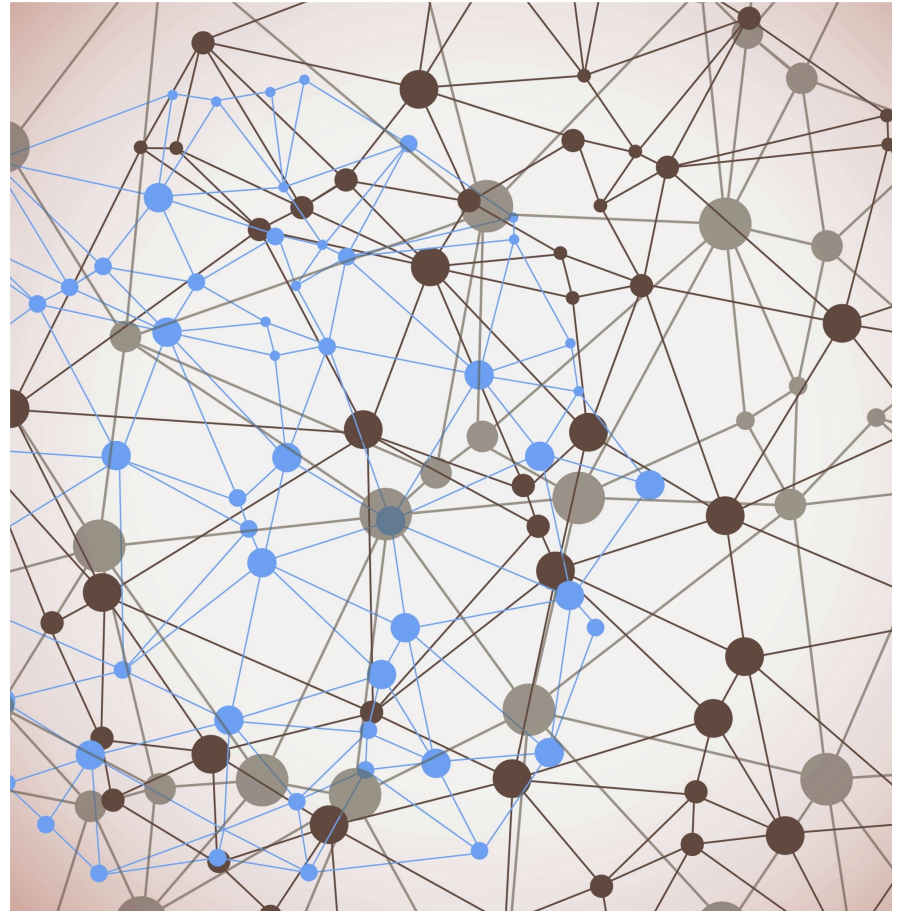

PROBABILITY THEORY



DATA DESCRIPTION

MEASURE OF CENTRAL TENDENCIES

- A **statistic** is a characteristic or measure obtained by using the data values from a sample.
 - A **parameter** is a characteristic or measure obtained by using all the data values from a specific population.
-

-
- There are many ways to organize and describe a data set.
 - Important characteristics to look for when organizing and describing a data set are its **center**, its **variability** (or spread), and its **shape**.
-

MEASURE OF CENTRAL TENDENCY

- A single value which represents the whole data set is called an **average** or **measure of location** or **measure of central tendency**.
 - It is a value that represents a typical, or central entry of a data set.
 - The three most commonly used measures of central tendency are the
 - mean,
 - median,
 - mode.
-

MEAN

The **mean** is the sum of the values divided by the total number of values. The symbol $\bar{X}$ represents the **sample mean**.

$$\bar{X} = \frac{X_1 + X_2 + X_3 + \cdots + X_n}{n} = \frac{\sum X}{n}$$

where n represents the total number of values in the sample. This is also known as **arithmetic mean**.

For **population mean**, the Greek letter μ (mu) is used for the mean.

$$\mu = \frac{X_1 + X_2 + X_3 + \cdots + X_N}{N} = \frac{\sum X}{N}$$

Where N represents the total number of values in the population.

EXAMPLE:

The data show the number of patients in a sample of six hospitals who acquired an infection while hospitalized. Find the mean.

110, 76, 29, 38, 105, 31

$$\bar{X} = \frac{110 + 76 + 29 + 38 + 105 + 31}{6}$$

$$= \frac{389}{6} = 64.8$$

The mean of the number of hospital infections for the six hospitals is 64.8.

ADVANTAGES OF MEAN:

- 1. Uses All Data Points:** It incorporates every value in the dataset, providing a comprehensive measure.
 - 2. Mathematically Stable:** It is easy to compute and widely used in statistical formulas.
 - 3. Useful for Further Analysis:** Essential for techniques like variance, standard deviation, and regression analysis.
-

DISADVANTAGES OF MEAN:

1. Sensitive to Outliers: Extreme values can distort the mean significantly.

Example: For {1,2,3,100}, the mean

$$\frac{1 + 2 + 3 + 100}{4} = 26.5$$

which doesn't represent the majority of the data.

2. May Not Represent Actual Values: In some cases, the mean may not correspond to any actual observation in the dataset.

MEAN OF A GROUPED DATA

The formula for the mean is

$$\bar{X} = \frac{\sum f \cdot X_m}{\sum f}$$

[Note: The symbols $\sum f \cdot X_m$ mean to find the sum of the product of the frequency (f) and the midpoint (X_m) for each class.]

MEAN OF A GROUPED DATA

Using the frequency distribution to find the mean of the data set of out-of-pocket prescription medicine expenses (in dollars) for 30 U.S. adults in a recent year.

Class	f	Midpoint X_m	$f \cdot X_m$
155-190	3	$\frac{155+190}{2} = 172.5$	517.5
191-226	2	$\frac{190+226}{2} = 208.5$	
227-262	5	$\frac{227+262}{2} = 244.5$	
263-298	6	$\frac{263+298}{2} = 280.5$	
299-334	7	$\frac{299+334}{2} = 316.5$	
335-370	4	$\frac{335+370}{2} = 352.5$	
371-406	3	$\frac{371+406}{2} = 388.5$	

MEDIAN

Definition: The **median** is the middle value of a dataset when the data is arranged in ascending or descending order. If the dataset has an even number of values, the median is the average of the two middle values.

Example:

1. Dataset (Odd number of elements):

$\{2, 4, 6, 8, 10\}$

Median = 6 (middle value).

2. Dataset (Even number of elements):

$\{2, 4, 6, 8\}$

Median = $\frac{4+6}{2} = \frac{10}{2} = 5$.

ADVANTAGES OF MEDIAN:

1. Insensitive to Outliers: Unlike the mean, the median is not affected by extreme values.

1. Example: For {1,2,3,100}, the median is

$$\frac{2 + 3}{2} = 2.5,$$

which better represents the dataset.

2. Easy to Compute: For small datasets, arranging the data and finding the middle value is straightforward.

DISADVANTAGES OF MEDIAN:

1. **Does Not Use All Data Points:** Only the middle value(s) is considered, ignoring the rest of the data.
 2. **Not Suitable for Mathematical Analysis:** Cannot be used in advanced statistical computations like variance or regression.
-

MODE

Definition: The **mode** is the most frequently occurring value(s) in a dataset. A dataset can have:

- One mode (**Unimodal**): A single value occurs most frequently.
 - Two modes (**Bimodal**): Two values occur with equal and maximum frequency. both values are considered to be the mode.
 - Many modes (**Multimodal**): More than two values occur with equal and maximum frequency. each value is used as the mode
 - No mode: When all values occur with the same frequency.
-

ADVANTAGES OF MODE:

- 1. Easy to Interpret:** It directly represents the most common value in a dataset.
 - 2. Useful for Categorical Data:** Can be applied to qualitative data (e.g., most preferred product or programming language).
 - 3. Unaffected by Extreme Values:** Outliers do not influence the mode.
-

DISADVANTAGES OF MODE:

1. **Limited to Frequency:** Only considers the most frequent value(s) and ignores other aspects of the data distribution.
 2. **May Not Exist or May Be Ambiguous:**
 1. Example: $\{1,2,3,4\}$ has no mode.
 2. Bimodal or multimodal distributions can complicate interpretation.
-

EXAMPLE:

At a political debate a sample of audience members was asked to name the political party to which they belong. Their responses are shown in the table. What is the mode of the responses?

Political Party	Frequency, f
Democrat	34
Republican	56
Other	21
Did not respond	9

MODE FOR GROUPED DATA

The mode for grouped data is the modal class. The **modal class** is the class with the largest frequency.

~~MODE OF A GROUPED DATA~~

Using the frequency distribution to find the modal class of the data set of out-of-pocket prescription medicine expenses (in dollars) for 30 U.S. adults in a recent year.

Class	f
155-190	3
191-226	2
227-262	5
263-298	6
299-334	7
335-370	4
371-406	3

~~MODE OF A GROUPED DATA~~

Using the frequency distribution to find the modal class of the data set of out-of-pocket prescription medicine expenses (in dollars) for 30 U.S. adults in a recent year.

Class	f
155-190	3
191-226	2
227-262	5
263-298	6
299-334	7
335-370	4
371-406	3

← Modal
class

COMPARING MEAN, MEDIAN AND MODE

Find the mean, median, and mode of the sample ages of a class shown. Which measure of central tendency best describes a typical entry of this data set? Are there any outliers?

Ages in a class						
20	20	20	20	20	20	21
21	21	21	22	22	22	23
23	23	23	24	24	65	

COMPARING MEAN, MEDIAN AND MODE

Find the mean, median, and mode of the sample ages of a class shown. Which measure of central tendency best describes a typical entry of this data set? Are there any outliers?

Ages in a class						
20	20	20	20	20	20	21
21	21	21	22	22	22	23
23	23	23	24	24	65	

$$\text{Mean} = \frac{20+20+\cdots+65}{20} \approx 23.8 \text{ years}$$

$$\text{Median} = \frac{21+22}{2} = 21.5 \text{ years}$$

$$\text{Mode} = 20 \text{ years}$$

WHEN TO USE EACH MEASURE

Use Mean:

- When the dataset is **symmetrical** and does not have extreme outliers.
- Example: Calculating the average download speed of an internet connection.

Use Median:

- When the dataset is **skewed** or has extreme outliers.
- Example: Analyzing median house prices in a city.

Use Mode:

- When analyzing **categorical** data or identifying the most frequent value in numerical data.
 - Example: Finding the most popular programming language among developers.
-

OTHER TYPES OF MEANS

Weighted Mean:

- Formula: Weighted Mean = $\frac{\sum w_i x_i}{\sum w_i}$
- Explanation: Incorporates weights based on importance or frequency of data points.
- Applications: Weighted averages in priority scheduling, weighted loss functions in machine learning.

Geometric Mean:

- Formula: Geometric Mean = $\sqrt[n]{x_1 \cdot x_2 \cdots x_n}$
 - Used for data involving ratios or growth rates (e.g., algorithm speed-ups, financial data in computer science).
-
- Applications: Evaluating average improvement factors over multiple test cases.

OTHER TYPES OF MEANS

Harmonic Mean:

- Formula: Harmonic Mean = $\frac{n}{\sum \frac{1}{x_i}}$
- Used when dealing with rates (e.g., throughput in computer networks, performance of parallel systems).
- Applications: Average time to complete tasks when multiple processors are involved.

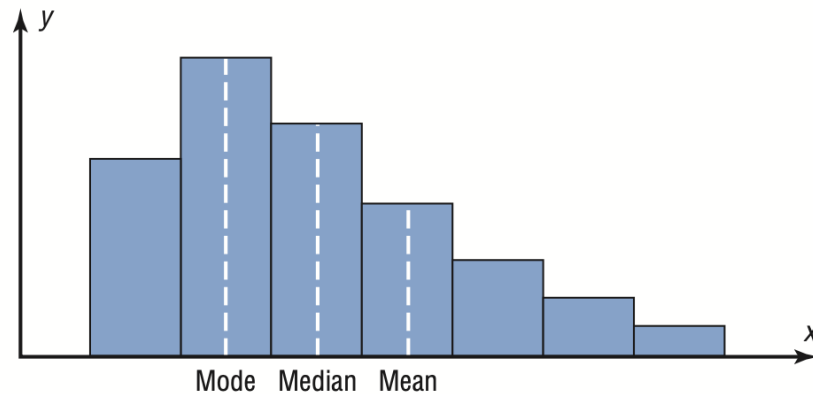
Trimmed Mean:

A trimmed mean is a measure of central tendency designed to reduce the influence of extreme values. This is achieved by discarding the smallest and largest k numbers

$$x_{trim} = \frac{\sum_{i=k+1}^{n-k} x_i}{n - 2k}$$

RELATIONSHIPS AMONG THE MEAN, MEDIAN AND MODE

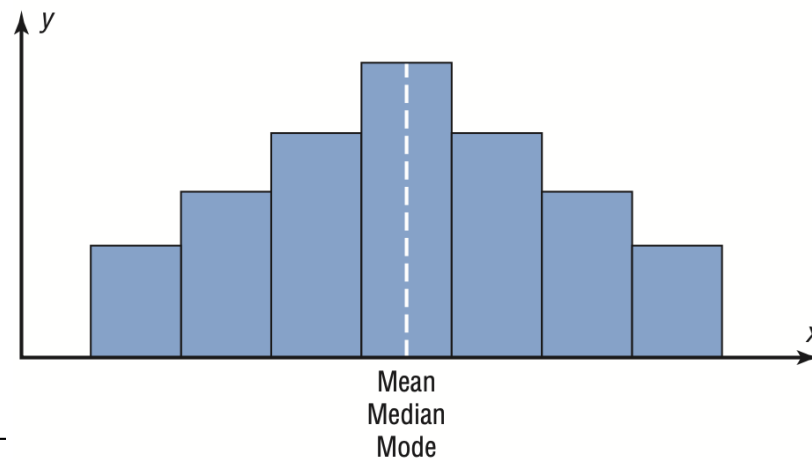
- Frequency distributions can assume many shapes.
- In a **positively skewed** or **right-skewed distribution**, the majority of the data values fall to the left of the mean and cluster at the lower end of the distribution; the “tail” is to the right. Also, the mean is to the right of the median, and the mode is to the left of the median.



(a) Positively skewed or right-skewed

RELATIONSHIPS AMONG THE MEAN, MEDIAN AND MODE

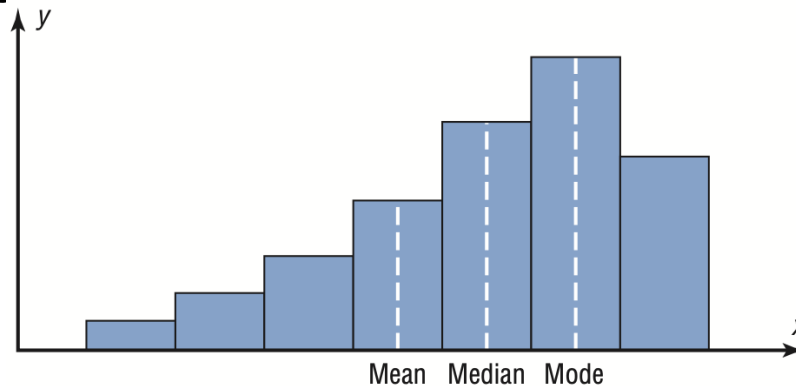
- In a **symmetric distribution**, the data values are evenly distributed on both sides of the mean. In addition, when the distribution is unimodal, the mean, median, and mode are the same and are at the center of the distribution.



(b) Symmetric

RELATIONSHIPS AMONG THE MEAN, MEDIAN AND MODE

- When the majority of the data values fall to the right of the mean and cluster at the upper end of the distribution, with the tail to the left, the distribution is said to be **negatively skewed** or **left-skewed**. Also, the mean is to the left of the median, and the mode is to the right of the median.



(c) Negatively skewed or left-skewed

MIDRANGE

The **midrange** is defined as the sum of the lowest and highest values in the data set, divided by 2.
The symbol MR is used for the midrange.

$$MR = \frac{\text{lowest value} + \text{highest value}}{2}$$

MEASURES OF VARIATION

Compare these two datasets:

D1: 12 13 13 14 15 14

D2: 5 9 12 15 20 20

They have identical means and medians but are very different

The numbers in D2 are more spread out. They have greater dispersion.

Such descriptive measures are referred to as **measures of variation** or **measures of spread** or **measures of dispersion**.

Dispersion is defined as the degree of spreadness or scatterness of observations from some average value (mean, median or mode) of the data.

MEASURES OF VARIATION

The **variance** is the average of the squares of the distance each value is from the mean. The symbol for the **population variance** is σ^2 (σ is the Greek lowercase letter sigma).

The formula for the population variance is

$$\sigma^2 = \frac{\sum (X_i - \mu)^2}{N}$$

Where,

X_i individual value

μ population mean

N population size

MEASURES OF VARIATION

The **standard deviation** is the square root of the variance. The symbol for the **population standard deviation** is σ .

The corresponding formula for the population standard deviation is

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{\sum (X_i - \mu)^2}{N}}$$

MEASURES OF VARIATION

The symbol for the **sample variance** is s .

The formula for the sample variance is

$$s^2 = \frac{\sum (X_i - \bar{X})^2}{n - 1}$$

Where,

X_i individual value

$\bar{X}$ sample mean

n sample size

MEASURES OF VARIATION

The symbol for the **sample standard deviation** is s .

The corresponding formula for the population standard deviation is

$$s = \sqrt{s^2} = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n - 1}}$$

SAMPLE VARIANCE AND STANDARD DEVIATION

Why the Difference in the Formulas? (The Role of $n - 1$)

1. Degrees of Freedom:

In the sample variance formula, the denominator $n - 1$ is the **degrees of freedom**. When we use the sample mean ($\bar{X}$) instead of the true population mean (μ), we lose one degree of freedom because the sample mean is calculated from the same data points. Therefore, dividing by n would underestimate the population variance.

2. Bias Correction:

If we divide by n in the sample variance formula, the sample variance tends to underestimate the population variance, especially for small sample sizes. Dividing by $n - 1$ corrects this bias and provides an **unbiased estimate** of the population variance.

Population variance uses all the data and divides by N because there is no uncertainty in the population mean.

Sample variance divides by $n-1$ to correct for the bias caused by estimating the population mean using the sample mean. This ensures the sample variance provides an unbiased estimate of the true population variance.

EXAMPLE:

Population Variance:

- Suppose you have the entire population data: {2,4,6,8}
- Population Mean (μ):

$$\mu = \frac{2 + 4 + 6 + 8}{4} = 5$$

- Population Variance (σ^2):

$$\sigma^2 = \frac{(2 - 5)^2 + (4 - 5)^2 + (6 - 5)^2 + (8 - 5)^2}{4} = \frac{9 + 1 + 1 + 9}{4} = 5$$

Sample Variance:

- Now, take a sample: {2,4,6}.
- Sample Mean ($\bar{x}$):

$$\bar{x} = \frac{2 + 4 + 6}{3} = 4$$

- Sample Variance (s^2):

$$s^2 = \frac{(2 - 4)^2 + (4 - 4)^2 + (6 - 4)^2}{3 - 1} = \frac{4 + 0 + 4}{2} = 4$$

SAMPLE VARIANCE AND STANDARD DEVIATION

When computing the variance for a sample, one might expect the following expression to be used:

$$\frac{\sum (X_i - \bar{X})^2}{n}$$

where $\bar{X}$ is the sample mean and n is the sample size.

This formula is not usually used, however, since in most cases the purpose of calculating the statistic is to estimate the corresponding parameter.

This expression does not give the best estimate of the population variance because it usually *underestimates* the population variance.

Therefore, instead of dividing by n , find the variance of the sample by dividing by $n - 1$, giving a slightly larger value and an unbiased estimate of the population variance.

SIMPLIFIED FORMULA FOR VARIATION AND STANDARD DEVIATION

Variance

$$\begin{aligned}s^2 &= \frac{\sum (X_i - \bar{X})^2}{n - 1} \\&= \frac{1}{n - 1} \left(\sum X_i^2 - \frac{(\sum X_i)^2}{n} \right) \\&= \frac{n \sum X_i^2 - (\sum X_i)^2}{n(n - 1)}\end{aligned}$$

Similarly, **standard deviation**

$$s = \sqrt{\frac{n \sum X_i^2 - (\sum X_i)^2}{n(n - 1)}}$$

EXAMPLE:

Find the sample variance and standard deviation for the amount of European auto sales for a sample of 6 years shown. The data are in millions of dollars.

11.2, 11.9, 12.0, 12.8, 13.4, 14.3

Solution

Find the sum of the values.

$$\Sigma X = 11.2 + 11.9 + 12.0 + 12.8 + 13.4 + 14.3 = 75.6$$

Square each value and find the sum.

$$\Sigma X^2 = 11.2^2 + 11.9^2 + 12.0^2 + 12.8^2 + 13.4^2 + 14.3^2 = 958.94$$

Substitute in the formulas and solve.

$$s^2 = \frac{n \Sigma X_i^2 - (\Sigma X_i)^2}{n(n-1)} = \frac{6(958.94) - (75.6)^2}{6(5)} = 1.276$$

The variance is 1.28 rounded.

$$s = \sqrt{1.28} = 1.13$$

Hence, the sample standard deviation is 1.13.

PROPERTIES OF VARIANCE AND STANDARD DEVIATION

Properties of Variance:

1. Variance of constant is zero. ie $V(c) = 0$
2. Variance is invariant to change of the origin. ie $V(X \pm a) = V(X)$
3. Variance is affected by change of scale. ie $V(aX) = a^2V(X)$

Properties of Standard Deviation

1. Standard Deviation of constant is zero. ie $S.D(c) = 0$
 2. Standard Deviation is invariant to change of the origin. ie $S.D(X \pm a) = S.D(X)$
 3. Standard Deviation is affected by change of scale. ie $S.D(aX) = |a| S.D(X)$
-

SAMPLE VARIANCE AND STANDARD DEVIATION FOR GROUPED DATA

The formula for the variance and standard deviation for grouped data is similar to that for finding the mean for grouped data, and it uses the midpoints X_m of each class.

$$\text{Variance} = s^2 = \frac{\sum f \cdot (X_m - \bar{X})^2}{n - 1}$$

$$= \frac{n \sum (f \cdot X_m^2) - (\sum f \cdot X_m)^2}{n(n - 1)}$$

EXAMPLE:

Find the variance and the standard deviation for the frequency distribution of the data. The data represent the number of miles that 20 runners ran during one week.

Class	Frequency	Midpoint
5.5–10.5	1	8
10.5–15.5	2	13
15.5–20.5	3	18
20.5–25.5	5	23
25.5–30.5	4	28
30.5–35.5	3	33
35.5–40.5	2	38

EXAMPLE:

Find the variance and the standard deviation for the frequency distribution of the data. The data represent the number of miles that 20 runners ran during one week.

A	B	C	D	E
Class	Frequency f	Midpoint X_m	$f \cdot X_m$	$f \cdot X_m^2$
5.5–10.5	1	8		
10.5–15.5	2	13		
15.5–20.5	3	18		
20.5–25.5	5	23		
25.5–30.5	4	28		
30.5–35.5	3	33		
35.5–40.5	2	38		

EXAMPLE:

Find the variance and the standard deviation for the frequency distribution of the data. The data represent the number of miles that 20 runners ran during one week.

A Class	B Frequency	C Midpoint	D $f \cdot X_m$	E $f \cdot X_m^2$
5.5–10.5	1	8	8	64
10.5–15.5	2	13	26	338
15.5–20.5	3	18	54	972
20.5–25.5	5	23	115	2,645
25.5–30.5	4	28	112	3,136
30.5–35.5	3	33	99	3,267
35.5–40.5	2	38	76	2,888
	<u>20</u>		<u>490</u>	<u>13,310</u>
	$n = 20$		$\Sigma f \cdot X_m = 490$	$\Sigma f \cdot X_m^2 = 13,310$

EXAMPLE:

Find the variance and the standard deviation for the frequency distribution of the data. The data represent the number of miles that 20 runners ran during one week.

A Class	B Frequency	C Midpoint	D $f \cdot X_m$	E $f \cdot X_m^2$	
5.5–10.5	1	8	8	64	$s^2 = 68.7$ $s = \sqrt{s^2} = 8.3$
10.5–15.5	2	13	26	338	
15.5–20.5	3	18	54	972	
20.5–25.5	5	23	115	2,645	
25.5–30.5	4	28	112	3,136	
30.5–35.5	3	33	99	3,267	
35.5–40.5	2	38	76	2,888	
	<u>20</u>		<u>490</u>	<u>13,310</u>	
	$n = 20$		$\Sigma f \cdot X_m = 490$	$\Sigma f \cdot X_m^2 = 13,310$	

INTERPRETING STANDARD DEVIATION

When interpreting the standard deviation, remember that it is a measure of the typical amount an entry deviates from the mean. The more the entries are spread out, the greater the standard deviation.

MEAN DEVIATION

Mean deviation, also known as the **average absolute deviation (AAD)**, is a measure of dispersion that shows how far the data points in a dataset deviate from the central tendency (typically the mean or median). It gives an idea of the **spread** or **variability** of the data by calculating the average of the **absolute deviations** from a central value.

MEAN DEVIATION

Formula for Mean Deviation

The formula depends on whether you calculate deviations from the **mean** or the **median**:

1. Mean Deviation from the Mean:

$$\text{Mean Deviation} = \frac{\sum |x_i - \bar{x}|}{n}$$

Where:

x_i : Individual data points.

$\bar{x}$: Mean of the dataset.

n : Number of data points.

2. Mean Deviation from the Median:

$$\text{Mean Deviation} = \frac{\sum |x_i - \text{Median}|}{n}$$

EXAMPLE:

Consider the dataset {2,4,6,8,10}

$\bar{x} = 6$ and median = 6

For mean:

$$\text{Mean Deviation} = \frac{\sum |x_i - \bar{x}|}{n} = \frac{|2 - 6| + |4 - 6| + |6 - 6| + |8 - 6| + |10 - 6|}{5} = 2.4$$

For median:

$$\text{Mean Deviation} = \frac{\sum |x_i - \text{median}|}{n} = \frac{|2 - 6| + |4 - 6| + |6 - 6| + |8 - 6| + |10 - 6|}{5} = 2.4$$

TYPES OF MEASURE OF DISPERSION

There are two types of measures of dispersion

1. Absolute Measure of Dispersion
 2. Relative Measure of Dispersion
-

TYPES OF MEASURE OF DISPERSION

Absolute Measure of Dispersion:

It is used to measure degree of scatterness in the data about some central point. These measures give result in same unit as the unit of data. Following are the types of absolute measure of dispersion Range,

Variance,

Standard Deviation,

Mean Deviation,

Quartile Deviation

Relative Measure of Dispersion:

A relative measure of dispersion is one that is expressed in the form of ratio, coefficient or percentage and is independent of units of measurement. It is useful for comparison of data of different nature. Following are the types of relative measure of dispersion

Co-efficient of Range,

Co-efficient of Variation,

Co-efficient of Standard Deviation,

Co-efficient of Mean Deviation,

Co-efficient of Quartile Deviation

Coefficient of Range:

Range is an absolute measure of dispersion. Its relative measure is known as co-efficient of dispersion and is defined as

$$\text{Coefficient of Range} = \frac{x_m - x_0}{x_m + x_0}$$

Coefficient of Variance:

Relative measure of variance is known as coefficient of variation, it is given by

$$\text{Coefficient of Variation} = C.V = \frac{s}{\bar{x}} 100$$

Coefficient of Standard Deviation:

Relative measure of standard deviation is known as co-efficient of variation, it is given by

$$\text{Coefficient of S.D} = \frac{s}{\text{Mean}}$$

Coefficient of Mean Deviation:

Relative measure of mean deviation is known as co-efficient of mean deviation, it is given by

$$\text{Coefficient of M.D} = \frac{M.D}{\text{Mean}}$$

Coefficient of Quartile Deviation:

Relative measure of quartile deviation is known as co-efficient of quartile deviation, it is given by

$$\text{Coefficient of Q.D} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

EXAMPLE:

The mean of the number of sales of cars over a 3-month period is 87, and the standard deviation is 5. The mean of the commissions is \$5225, and the standard deviation is \$773. Compare the variations of the two.

Solution:

The coefficients of variation are

For Sales

$$CVar = \frac{5}{87}100 = 5.7\%$$

For Commissions

$$CVar = \frac{773}{5225}100 = 14.8\%$$

Since the coefficient of variation is larger for commissions, the commissions are more variable than the sales.

A RANGE RULE OF THUMB

A rough estimate of the standard deviation is

$$s \sim \frac{\text{range}}{4}$$

In other words, if the range is divided by 4, an approximate value for the standard deviation is obtained.

A RANGE RULE OF THUMB

For example, the standard deviation for the data set 5, 8, 8, 9, 10, 12, and 13 is 2.7, and the range is $13-5=8$.

The range rule of thumb is $s \sim 2$.

The range rule of thumb in this case underestimates the standard deviation somewhat; however, it is in the ballpark.

CHEBYSHEV'S THEOREM

The proportion of values from a data set that will fall within k standard deviations of the mean will be at least

$$1 - \frac{1}{k^2},$$

where k is a number greater than 1 (k is not necessarily an integer).

CHEBYSHEV'S THEOREM

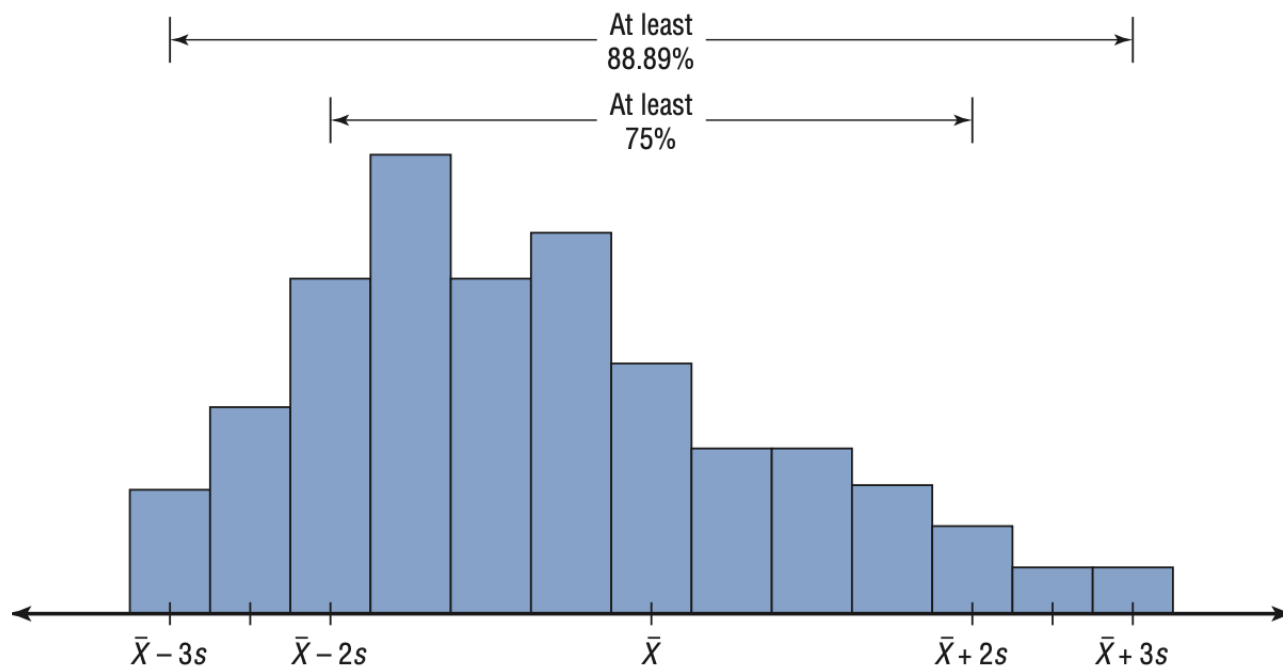
This theorem states that at least three-fourths, or 75%, of the data values will fall within 2 standard deviations of the mean of the data set. This result is found by substituting $k = 2$ in the expression.

$$1 - \frac{1}{k^2} = 1 - \frac{1}{2^2} = 1 - \frac{1}{4} = \frac{3}{4} \quad \text{or} \quad 75\%$$

Furthermore, the theorem states that at least eight-ninths, or 88.89%, of the data values will fall within 3 standard deviations of the mean. This result is found by letting $k = 3$ and substituting in the expression.

$$1 - \frac{1}{k^2} = 1 - \frac{1}{3^2} = 1 - \frac{1}{9} = \frac{8}{9} \quad \text{or} \quad 88.89\%$$

CHEBYSHEV'S THEOREM



EXAMPLE:

For example, if a variable has a mean of 70 and a standard deviation of 1.5, at least three-fourths, or 75%, of the data values fall between 67 and 73. These values are found by adding 2 standard deviations to the mean and subtracting 2 standard deviations from the mean, as shown:

$$70 + 2(1.5) = 70 + 3 = 73 \quad \text{and} \quad 70 - 2(1.5) = 70 - 3 = 67$$

And at least eight-ninths, or 88.89%, of the data values fall between 65.5 and 74.5, since

$$70 + 3(1.5) = 70 + 4.5 = 74.5 \quad \text{and} \quad 70 - 3(1.5) = 70 - 4.5 = 65.5$$

EXAMPLE:

The average sale price of new one-family houses in the United States for 2003 was \$246,300. Find the range of values in which at least 75% of the sale prices will lie if the standard deviation is \$48,500.

75% means two standard deviation of the mean. Since 48,500 is the standard deviation and 246,300 is the mean. The range will be

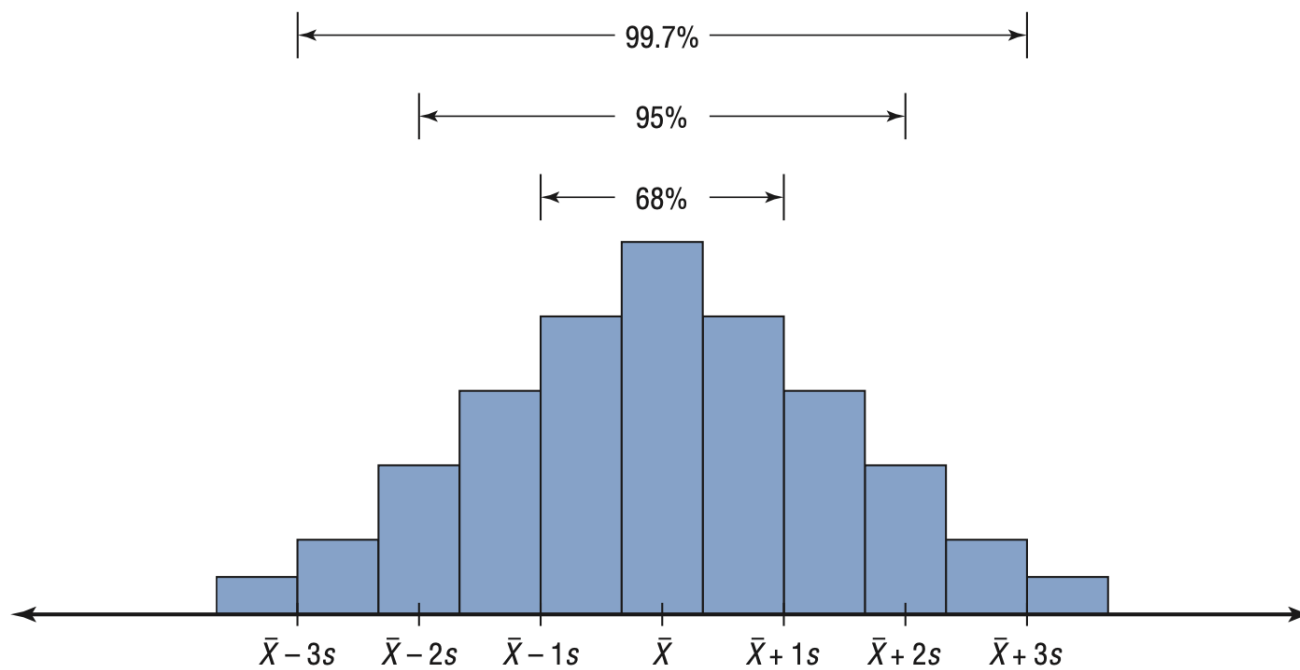
From $246,300 - 2(48,500) = \$149,300$

To $246,300 + 2(48,500) = \$343,300$

THE EMPIRICAL (NORMAL) RULE

When a distribution is **bell-shaped** (or what is called normal), the following statements, which make up the empirical rule, are true.

- Approximately 68% of the data values will fall within 1 standard deviation of the mean.
 - Approximately 95% of the data values will fall within 2 standard deviations of the mean.
 - Approximately 99.7% of the data values will fall within 3 standard deviations of the mean.
-



PROBABILITY

Probability is a number assigned by certain rules such that it describes the relative size of the event E compared with the sample space Ω .

So, when we say that probability is a measure of the size of a set, we create a mapping that takes in a set and outputs the size of that set.

PROBABILITY SPACE $(\Omega, \mathcal{F}, P)$:

Our discussion will be based on the slogan **probability is a measure of the size of a set**.

Three elements constitute a probability space:

Sample Space Ω : The set of all possible outcomes from an experiment.

Event Space $\mathcal{F}$: The collection of all possible events. An event E is a subset in Ω that defines an outcome or a combination of outcomes. which must be a σ -algebra

Probability Law P : A mapping from an event E to a number $P[E]$ which, ideally, measures the size of the event. P is also known as **probability measure**.

Therefore, whenever you talk about “probability,” you need to specify the triplet $(\Omega, \mathcal{F}, P)$ to define the probability space.

TYPES OF SAMPLE SPACE:

A sample space can contain discrete outcomes or continuous outcomes.

Discrete Sample Space:

- Coin flip: $\Omega = \{H, T\}$
- Rock/Paper/Scissor: $\Omega = \{rock, paper, scissor\}$
- Even integer: $\Omega = \{2, 4, 6, \dots\}$

Continuous Sample Space:

- Waiting time for a bus: $\Omega = \{t | 0 \leq t \leq 30 \text{ minutes}\}$
- Frequency of a pitch: $\Omega = \{f | 0 \leq f \leq f_{max}\}$

Functional Sample Space:

- Set of straight lines in 2D: $\Omega = \{f | f(x) = ax + b, a, b \in \mathbb{R}\}$
 - Set of cosine functions: $\Omega = \{f | f(x) = \cos(2\pi\omega_0 t + \theta) \mid 0 \leq \theta \leq 2\pi\}$
-

EVENT:

- An **event** consists of a set of outcomes of a probability experiment.
 - An event can be one outcome or more than one outcome.
 - An event with one outcome is called a **simple event**.
 - An event with more than one outcome is called a **compound event**.
 - In general, a compound event consists of two or more outcomes or simple events.
-

EVENT:

- **Impossible Event:** An event that contains no sample point is called impossible event. It is also called null event.
 - **Sure Event:** An event that contains all the sample points is called sure event. It is also called certain event.
 - **Mutually Exclusive Event:** Two events A and B are said to be mutually exclusive if they cannot occur at the same time ie $A \cap B = \phi$.
 - For example: When a coin is tossed once then the events $A = \{H\}$ and $B = \{T\}$ are mutually exclusive events.
 - **Collectively Exhaustive Events:** Two events A and B are said to be collectively exhaustive events if their union is sample space itself ie $A \cup B = S$.
 - **Equally Likely Events:** Two events A and b are said to be equally likely, if number of elements in A and B are same. ie $n(A) = n(B)$.
-

PROBABILITY

- **Probability** as a general concept can be defined as the chance of an event occurring.
 - Probability is between 0 and 1.
 - 0 means it will never happen and 1 means it will most certainly happen.
 - A **probability experiment** is an action, or trial, through which specific results (counts, measurements, or responses) are obtained.
-

σ -ALGEBRA

The term σ -algebra is used in measure theory to denote a collection of sets that satisfy certain conditions listed below.

It is a way of stating that if we assign a probability to certain events, we also need to assign a probability to their complements and to their union.

If our sample space is discrete, a possible choice for the σ -algebra is the power set of the sample space, which consists of all possible sets of elements in the sample space.

σ -ALGEBRA

A **σ -algebra** $\mathcal{F}$ is a collection of sets in Ω such that:

1. $\phi, \Omega \in \mathcal{F}$
 2. If a set $S \in \mathcal{F}$ then $S^c \in \mathcal{F}$.
 3. If the sets $S_1, S_2 \in \mathcal{F}$, then $S_1 \cup S_2 \in \mathcal{F}$. This also holds for infinite sequences; if $S_1, S_2, \dots \in \mathcal{F}$ then $\bigcup_{i=1}^{\infty} S_i \in \mathcal{F}$.
 4. If the sets $S_1, S_2 \in \mathcal{F}$, then $S_1 \cap S_2 \in \mathcal{F}$. This also holds for infinite sequences; if $S_1, S_2, \dots \in \mathcal{F}$ then $\bigcap_{i=1}^{\infty} S_i \in \mathcal{F}$.
-

PROBABILITY SPACE $(\Omega, \mathcal{F}, P)$:

- If we do not specify the sample space Ω , then there is no way to define the events.
 - If we do not have an event space $\mathcal{F}$ which is a σ -algebra, then some events will become undefined, and further, if the probability law is applied only to outcomes, we will not be able to define the probability for events.
 - Finally, if we do not specify the probability law, then we do not have a way to assign probabilities.
-

EXAMPLE:

A doctor is interested in modeling the cholesterol levels of her patients probabilistically. Every time a patient visits her, she tests their cholesterol level.

Here the experiment is the cholesterol test, the outcome is the measured cholesterol level, and the sample space Ω is the positive real line.

The doctor is mainly interested in whether the patients to have low, borderline-high, or high cholesterol.

- The event L (low cholesterol) contains all outcomes below 200 mg/dL,
 - the event B (borderline-high cholesterol) contains all outcomes between 200 and 240 mg/dL, and
 - the event H (high cholesterol) contains all outcomes above 240 mg/dL.
- The σ -algebra $\mathcal{F}$ of possible events therefore equals

$$\mathcal{F} := \{L \cup B \cup H, L \cup B, L \cup H, B \cup H, L, B, H, \emptyset\}.$$

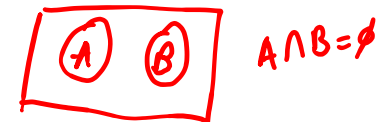
The events are a partition of the sample space, which simplifies deriving the corresponding σ -algebra.

AXIOMS OF PROBABILITY



A probability law is a function $P: \mathcal{F} \rightarrow [0, 1]$ that maps an event A to a real number in $[0, 1]$. The function must satisfy the axioms of probability:

- I. Non-negativity: $P[\underline{A}] \geq 0$, for any $A \subseteq \Omega$.
- II. Normalization: $P[\underline{\Omega}] = 1$.
- III. Additivity: For any disjoint sets $\{A_1, A_2, \dots\}$, it must be true that
$$P \left[\bigcup_{i=1}^{\infty} A_i \right] = \sum_{i=1}^{\infty} P[A_i].$$



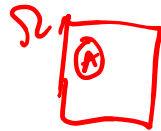
$$P(A \cup B) = P(A) + P(B)$$

COROLLARY:

Let $A \in \mathcal{F}$ be an event. Then,

- a) $P[A^c] = 1 - P[A]$. ✓
- b) If $A \subset B$, $P[A] \leq P[B]$
- c) $P[A] \leq 1$.
- d) $P[\emptyset] = 0$.

$$A \subset \Omega \\ P(A) \leq P(\Omega) = 1$$

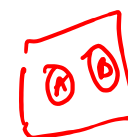


$$\begin{aligned} & P(A) \\ & A^c, A \quad A^c \cap A = \emptyset, \quad A^c \cup A = \Omega \\ & 1 = P(\Omega) = P(A^c \cup A) = P(A^c) + P(A) \\ & \Rightarrow P(A^c) + P(A) = 1 \\ & P(A^c) = 1 - P(A) \end{aligned}$$

COROLLARY: (UNION OF TWO NON-DISJOINT SETS)

Let $A, B \in \mathcal{F}$ be events,

$$P[A \cup B] = P[A] + P[B] - P[A \cap B]$$



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

PROVE THAT

$$P(\emptyset) = 0$$

$$P(A^c) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P[A^c \cap B^c] = 1 + P[A \cap B] - P[A] - P[B]$$

Hint: Use De Morgan's Law that

$$A^c \cap B^c = (A \cup B)^c$$

$$P(A^c \cap B^c) = P(A \cup B)^c = 1 - P(A \cup B) = 1 - P(A) - P(B) + P(A \cap B)$$

EXERCISE:

Let the events A and B have $P[A] = x$, $P[B] = y$ and $P[A \cup B] = z$. Find the following probabilities:

$$P[A \cap B],$$

$$P[A^c \cup B^c], \text{ and}$$

$$P[A \cap B^c].$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$z = x + y - P(A \cap B) \Rightarrow P(A \cap B) = x + y - z$$

EXAMPLE:

$$\frac{6}{\text{Total}} = \frac{6}{6 \times 6} = \frac{6}{36}$$

$$P(E) = \frac{\text{no. of } E}{\text{Total no.}}$$

$$A = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$$

$$B = \{(6,5), (5,6)\}$$

What is the probability of getting a total of 7 or 11 when a pair of fair dice is tossed?

Solution:

$$P(A \cup B) = P(A) + P(B)$$

Let A be the event that 7 occurs and B the event that 11 comes up. Now, a total of 7 occurs for 6 of the 36 sample points, and a total of 11 occurs for only 2 of the sample points. Since all sample points are equally likely, we have $P(A) = 1/6$ and $P(B) = 1/18$. The events A and B are mutually exclusive, since a total of 7 and 11 cannot both occur on the same toss. Therefore,

$$P(A \cup B) = P(A) + P(B) = \frac{1}{6} + \frac{1}{18} = \frac{2}{9}$$

CONDITIONAL PROBABILITY

Consider a probabilistic space $(\Omega, \mathcal{F}, P)$ where we find out that the outcome of the experiment belongs to a certain event $S \in \mathcal{F}$.

This obviously affects how likely it is for any other event $S' \in \mathcal{F}$ to have occurred:

We can rule out any outcome not belonging to S . The updated probability of each event is known as the **conditional probability** of S' **given** S . $P(S'|S)$

Intuitively, the conditional probability can be interpreted as the fraction of outcomes in S that are also in S' ,

$$P(S'|S) = \frac{\text{outcomes in } \underline{S'} \text{ and } \underline{S}}{\text{outcomes in } S}$$

$$= \frac{P(S' \cap S)}{P(S)}$$

CONDITIONAL PROBABILITY

We have

$$P(S|S) = 1$$

and

if S and S' are disjoint then

$$P(\underline{S'}|S) = 0.$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\frac{P(S \cap S)}{P(S)} = \frac{P(S)}{P(S)} = 1$$

CONDITIONAL PROBABILITY

The conditional probability $P(\cdot | S)$ is a valid probability measure in the probability space $(S, \mathcal{F}_S, P(\cdot | S))$, where $\mathcal{F}_S$ is a σ -algebra that contains the intersection of S and the sets in $\mathcal{F}$.

CONDITIONAL PROBABILITY

when we condition on an intersection of sets we write the conditional probability as

$$\underline{P(S|A, B, C)} := P(S|A \cap B \cap C),$$

for any events S, A, B, C .

EXAMPLE:

JFK airport hires you to estimate how the punctuality of flight arrivals is affected by the weather. You begin by defining a probability space for which the sample space is

$\Omega = \{\text{late and rain, late and no rain, on time and rain, on time and no rain}\}$

and the σ -algebra is the power set of Ω . From data of past flights you determine that a reasonable estimate for the probability measure of the probability space is

$$\begin{aligned} P(\text{late, no rain}) &= \frac{2}{20} & P(\text{late, rain}) &= \frac{3}{20} \\ P(\text{on time, no rain}) &= \frac{14}{20} & P(\text{on time, rain}) &= \frac{1}{20} \end{aligned}$$

EXAMPLE: $R = (R \cap L) \cup (R \cap L^c)$
 $P(R) = P[(R \cap L) \cup (R \cap L^c)] = P(R \cap L) + P(R \cap L^c)$



$R \cap L$
 $R \cap L^c$

JFK airport hires you to estimate how the punctuality of flight arrivals is affected by the weather. You begin by defining a probability space for which the sample space is

$\Omega = \{\text{late and rain, late and no rain, on time and rain, on time and no rain}\}$

$$P(\text{late, no rain}) = \frac{2}{20}$$

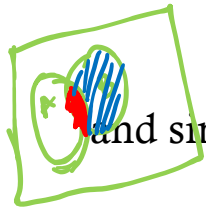
$$P(\overline{\text{late}}, \text{rain}) = \frac{3}{20}$$

$$P(\text{on time, no rain}) = \frac{14}{20}$$

$$P(\text{on time}, \text{rain}) = \frac{1}{20}$$

$$P(\text{late} | \text{rain}) = \frac{P(\text{late, rain})}{P(\text{rain})} = \frac{\left(\frac{3}{20}\right)}{\left(\frac{3}{20} + \frac{1}{20}\right)} = \frac{3}{4}$$

$$\begin{aligned} P(\text{late} | \text{rain}) &= \frac{P(\text{late and rain})}{P(\text{rain})} \\ &= \frac{\frac{3}{20}}{\left(\frac{3}{20} + \frac{1}{20}\right)} \\ &= \frac{\frac{3}{20}}{\frac{4}{20}} = \frac{3}{4} \end{aligned}$$



$$B = (B \cap A) \cup (B \cap A^c)$$

and similarly $P(\text{late} | \text{no rain}) = 1/8$

$$P(B) = P(B \cap A) + P(B \cap A^c)$$

EXAMPLE:

Suppose that our sample space S is the population of adults in a small town who have completed the requirements for a college degree. We shall categorize them according to gender and employment status. The data are given in table

	Employed	Unemployed	Total
Male	460	40	500
Female	140	260	400
Total	600	300	900

One of these individuals is to be selected at random for a tour throughout the country to publicize the advantages of establishing new industries in the town. What is the probability of selecting an employed who is a male?

EXAMPLE:

	Employed	Unemployed	Total
Male	460	40	500
Female	140	260	400
Total	600	300	900

One of these individuals is to be selected at random for a tour throughout the country to publicize the advantages of establishing new industries in the town. What is the probability of selecting an employed who is a male?

$$\begin{aligned}
 P(M|E) &= \frac{n(M \cap E)}{n(E)} = \frac{P(M \cap E)}{P(E)} = \frac{n(M \cap E)/n(S)}{n(E)/n(S)} \\
 &= \frac{460}{900} \cdot \frac{900}{600} = \frac{23}{30}
 \end{aligned}$$

$$\begin{aligned}
 P(Male|Emp) &= \frac{P(Male \cap Emp)}{P(Emp)} \\
 &= \frac{460/900}{600/900} = \frac{460}{600} = \frac{23}{30}
 \end{aligned}$$

EXAMPLE

	Employed	Unemployed	Total
Male	460	40	500
Female	140	260	400
Total	600	300	900

What is the probability of selecting a male who is employed?

P(Employed given Male)

$$\begin{aligned}P(E|M) &= \frac{n(E \cap M)}{n(M)} \\&= \frac{460}{500} = \frac{23}{25}\end{aligned}$$

NOTE:

Since

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

We get

$$P(A \cap B) = P(B)P(A|B)$$

EXAMPLE:

A box contains 15 items, 4 of which are defective and 11 are good. Two items are selected. What is the probability that the *first* is good and the *second* defective.

Let A : First item selected is good, B : Second item selected is defective.

We need to find $P(A \cap B)$. But,

$$P(A \cap B) = P(A)P(B|A)$$

$$P(B|A) =$$

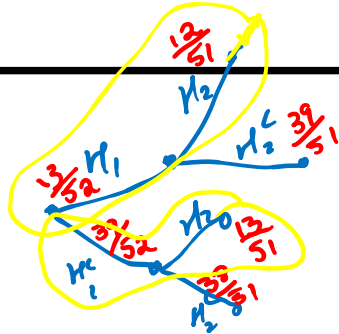
Clearly, $P(A) = \frac{11}{15}$

Given that event A has already occurred, there remains 14 items of which 4 are defective. Therefore,

$$P(B|A) = \frac{4}{14}$$

Hence, $P(A \cap B) = P(A)P(B|A) = \frac{11}{15} \cdot \frac{4}{14} = 0.16$

EXAMPLE:



Two cards are dealt from a pack of ordinary playing cards. Find the probability that the second card is a heart.

$$H_2 = (H_1 \cap H_2) \cup (H_1^c \cap H_2)$$

H_1 : First card is a heart, H_2 : Second card is Heart, $P(H_2) = P(H_1 \cap H_2) + P(H_1^c \cap H_2)$

$$\begin{aligned} P(H_2) &= P(H_1 \cap H_2) + P(H_1^c \cap H_2) \\ &= P(H_1)P(H_2|H_1) + P(H_1^c)P(H_2|H_1^c) \\ &= \left(\frac{13}{52}\right)\left(\frac{12}{51}\right) + \left(\frac{39}{52}\right)\left(\frac{13}{51}\right) = \frac{1}{4} \end{aligned}$$

THEOREM:

Let $(\Omega, \mathcal{F}, P)$ be a probability space and $S_1, S_2, \dots$ a collection of events in $\mathcal{F}$,

$$\begin{aligned} P(\cap_i S_i) &= P(S_1)P(S_2|S_1)P(S_3|S_1 \cap S_2) \dots \\ &= \prod_i P(S_i | \cap_{j=1}^{i-1} S_j). \end{aligned}$$

$$P(S_1 \cap S_2 \cap S_3 \cap S_4) = P(S_1) P(S_2|S_1) P(S_3|S_1 \cap S_2) P(S_4|S_1 \cap S_2 \cap S_3)$$

EXAMPLE:

Three cards are drawn in succession, without replacement, from an ordinary deck of playing cards. Find the probability that the event $A_1 \cap A_2 \cap A_3$ occurs, where A_1 is the event that the first card is a red ace, A_2 is the event that the second card is a 10 or a jack, and A_3 is the event that the third card is greater than 3 but less than 7.

Solution: $P(A_1) = \frac{2}{52}, P(A_2|A_1) = \frac{8}{51}, P(A_3|A_1 \cap A_2) = \frac{12}{50}$

and hence, by the theorem,

$$\begin{aligned} P(A_1 \cap A_2 \cap A_3) &= P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \\ &= \frac{2}{52} \frac{8}{51} \frac{12}{50} = \frac{8}{5525}. \end{aligned}$$

$$P(X|Y) = \frac{P(X \cap Y)}{P(Y)} \quad P(X \cap Y) = P(Y) P(X|Y)$$

THEOREM: (LAW OF TOTAL PROBABILITY).

$$S = (S \cap A_1) \cup (S \cap A_2) \cup (S \cap A_3) \cup \dots \cup (S \cap A_n)$$

$$P(S) = P(S \cap A_1) + P(S \cap A_2) + P(S \cap A_3) + \dots + P(S \cap A_n)$$

Let $(\Omega, \mathcal{F}, P)$ be a probability space and let the collection of disjoint sets $A_1, A_2, \dots, A_n \in \mathcal{F}$ be any partition of Ω . For any set $S \in \mathcal{F}$

$$= P(A_1) P(S|A_1) + P(A_2) P(S|A_2) + \dots + P(A_n) P(S|A_n) \quad P(S)$$

$$P(S) = \sum_i P(S \cap A_i)$$

$$= \sum_i P(A_i) P(S|A_i) \quad \checkmark$$

$$S = (S \cap A) \cup (S \cap A^c)$$

$$\begin{aligned} P(S) &= P(S \cap A) + P(S \cap A^c) \\ &= P(A) P(S|A) + P(A^c) P(S|A^c) \end{aligned}$$

EXAMPLE:

Your aunt is arriving at JFK tomorrow and you would like to know how likely it is for her flight to be on time. From previous example, you recall that

$$P(\text{late} | \text{rain}) = 0.75, P(\text{late} | \text{no rain}) = 0.125.$$

$$P(\text{Late}) = P(\text{rain}) P(\text{late} | \text{rain}) + P(\text{no rain}) P(\text{late} | \text{no rain})$$

After checking out a weather website, you determine that $P(\text{rain}) = 0.2$.

Now, how can we integrate all of this information?

The events rain and no rain are disjoint and cover the whole sample space, so they form a partition. We can consequently apply the law of total probability to determine

$$\begin{aligned} P(\text{late}) &= P(\text{late} | \text{rain}) P(\text{rain}) + P(\text{late} | \text{no rain}) P(\text{no rain}) \\ &= 0.75 \cdot 0.2 + 0.125 \cdot 0.8 = 0.25. \end{aligned}$$

So the probability that your aunt's plane is late is $1/4$.

EXAMPLE:

In a certain assembly plant, three machines, B1, B2, and B3, make 30%, 45%, and 25%, respectively, of the products. It is known from past experience that 2%, 3%, and 2% of the products made by each machine, respectively, are defective. Now, suppose that a finished product is randomly selected. What is the probability that it is defective?

EXAMPLE: (SOLUTION)

Consider the following events:

A: the product is defective,

B1: the product is made by machine B1,

B2: the product is made by machine B2,

B3: the product is made by machine B3.

Applying the rule of elimination, we can write

$$P(A) = P(B1)P(A|B1) + P(B2)P(A|B2) + P(B3)P(A|B3).$$

Referring to the tree diagram, we find that the three branches give the probabilities

$$P(B1)P(A|B1) = (0.3)(0.02) = 0.006,$$

$$P(B2)P(A|B2) = (0.45)(0.03) = 0.0135,$$

$$P(B3)P(A|B3) = (0.25)(0.02) = 0.005,$$

Therefore.

$$P(A) = 0.006 + 0.0135 + 0.005 = 0.0245.$$

NOTE THAT:

$$\begin{aligned} P(A \cap B) &= P(A)P(B|A) \\ &= P(B)P(A|B). \end{aligned}$$

BAYES' THEOREM

For any two events A and B a probability space $(\Omega, \mathcal{F}, P)$ such that $P[A] > 0$ and $P[B] > 0$,

$$P(A)P(B|A) = P(B)P(A|B)$$

Bayes' theorem provides two views of the intersection $P[A \cap B]$ using two different conditional probabilities. We call $P[B|A]$ the **conditional probability** and $P[A|B]$ the **posterior probability**.

EXAMPLE:

You explain the probabilistic model described earlier to your cousin Marvin who lives in California.

A day later, you tell him that your aunt arrived late but you don't mention whether it rained or not.

After he hangs up, Marvin wants to figure out the probability that it rained.

Recall that the probability of rain was 0.2, but since your aunt arrived late he should update the estimate.

EXAMPLE:

Applying Bayes' rule and the law of total probability:

$$\begin{aligned} P(\text{rain}|\text{late}) &= \frac{P(\text{late}|\text{rain})P(\text{rain})}{P(\text{late})} \\ &= \frac{P(\text{late}|\text{rain})P(\text{rain})}{P(\text{late}|\text{rain})P(\text{rain}) + P(\text{late}|\text{no rain})P(\text{no rain})} \\ &= \frac{0.75 \cdot 0.2}{0.75 \cdot 0.2 + 0.125 \cdot 0.8} = 0.6 \end{aligned}$$

As expected, the probability that it rained increases under the assumption that your aunt is late.

EXAMPLE:

A certain disease affects 1% of a population. A test for this disease has:

True positive rate (sensitivity): 90%

False positive rate: 5%

If a randomly selected person tests positive, what is the probability that they actually have the disease?

EXAMPLE: (SOLUTION)

Given Data

$$P(D) = 0.01 \rightarrow P(\text{not } D) = 1 - 0.01 = 0.99$$

$$\text{True positive rate (sensitivity): } P(+ | D) = 0.90 \quad \text{False positive rate: } P(+ | \text{not } D) = 0.05$$

We are asked to find: $P(D | +) = ?$

Step 1: Use Bayes' Theorem

$$P(D | +) = \frac{P(+ | D)P(D)}{P(+ | D)P(D) + P(+ | \text{not } D)P(\text{not } D)}$$

Step 2: Substitute values

$$P(D | +) = \frac{0.90(0.01)}{0.90(0.01) + 0.05(0.99)}$$

Step 3: Compute final probability

$$P(D | +) = \frac{0.009}{0.0585} = 0.1538$$

Final Answer:

$$\boxed{P(D | +) \approx 0.154}$$

GENERALIZED BAYE'S THEOREM

If the events $B_1, B_2, \dots, B_k$ constitute a partition of the sample space S and $P(B_i) \neq 0$ for $i = 1, 2, \dots, k$ then for any event A in S

$$P(B_j|A) = \frac{P(B_j)P(A|B_j)}{\sum_{i=1}^k P(A|B_i)P(B_i)}$$

for $j = 1, 2, \dots, k$

INDEPENDENT EVENTS:

Now consider an experiment in which 2 cards are drawn in succession from an ordinary deck, with replacement. The events are defined as

A: the first card is an ace,

B: the second card is a spade.

Since the first card is replaced, our sample space for both the first and the second

draw consists of 52 cards, containing 4 aces and 13 spades. Hence, $P(B|A) = 1/4$ and $P(B) = 13/52 = 1/4$.

That is, $P(B|A) = P(B)$. When this is true, the events A and B are said to be independent.

INDEPENDENT EVENTS:

Conditional probabilities quantify the extent to which the knowledge of the occurrence of a certain event affects the probability of another event.

In some cases, it makes no difference: the events are independent.

More formally, events A and B are independent if and only if

$$P(A|B) = P(A)$$

Note: This definition is not valid if $P(B) = 0$.

INDEPENDENT EVENTS:

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Two events $A, B \in \mathcal{F}$ are **independent** if and only if

$$P[A \cap B] = P[A]P[B].$$

EXAMPLE:

A small town has one fire engine and one ambulance available for emergencies. The probability that the fire engine is available when needed is 0.98, and the probability that the ambulance is available when called is 0.92. In the event of an injury resulting from a burning building, find the probability that both the ambulance and the fire engine will be available, assuming they operate independently.

Solution : Let A and B represent the respective events that the fire engine and the ambulance are available. Then

$$P(A \cap B) = P(A)P(B) = (0.98)(0.92) = 0.9016.$$

NAÏVE BAYES CLASSIFICATION

Naïve Bayes applies Bayes' theorem to classification. Given a feature vector $x = (x_1, \dots, x_d)$ and classes $C = \{c_1, \dots, c_k\}$, Bayes gives:

$$P(c_j | x) = \frac{P(x | c_j) P(c_j)}{P(x)}.$$

Naïve Bayes assumption: features are conditionally independent given the class. So

$$P(x | c_j) = \prod_{i=1}^d P(x_i | c_j).$$

Thus, the posterior is

$$P(c_j | x) = P(c_j) \prod_{i=1}^d P(x_i | c_j),$$

and we pick the class with largest $P(c_j | x)$.

EXAMPLE:

- Suppose we have data describing whether people **play tennis** based on two features:
- **Weather** = {Sunny, Rainy}
- **Temperature** = {Hot, Mild}
- **Class (Target):** Play = {Yes, No}

Day	Weather	Temperature	Play
1	Sunny	Hot	No
2	Sunny	Mild	Yes
3	Rainy	Mild	Yes
4	Rainy	Hot	No
5	Sunny	Mild	Yes

SOLUTION:

Step 1 — Compute Priors

$$P(Yes) = \frac{3}{5} = 0.6, P(No) = \frac{2}{5} = 0.4$$

Step 2 — Compute Likelihoods

Compute probabilities of each feature value **given** the class.

For Play = Yes:

$$\begin{aligned} P(Sunny \mid Yes) &= \frac{2}{3}, & P(Rainy \mid Yes) &= \frac{1}{3} \\ P(Hot \mid Yes) &= \frac{0}{3} = 0, & P(Mild \mid Yes) &= \frac{3}{3} = 1 \end{aligned}$$

For Play = No:

$$\begin{aligned} P(Sunny \mid No) &= \frac{1}{2}, & P(Rainy \mid No) &= \frac{1}{2} \\ P(Hot \mid No) &= \frac{2}{2} = 1, & P(Mild \mid No) &= \frac{0}{2} = 0 \end{aligned}$$

SOLUTION:

Predict for a New Case

New day:

Weather = Sunny, Temperature = Mild $\rightarrow$ Will they play?

We compute both posteriors.

- **For Play = Yes:**

- $P(Yes | Sunny, Mild) \propto P(Yes) \times P(Sunny | Yes) \times P(Mild | Yes)$

- $= 0.6 \times \frac{2}{3} \times 1 = 0.6 \times 0.6667 = 0.4$

- **For Play = No:**

- $P(No | Sunny, Mild) \propto P(No) \times P(Sunny | No) \times P(Mild | No)$

- $= 0.4 \times \frac{1}{2} \times 0 = 0$

VARIABLES:

Let's say we want to define a variable of height of a person.

It has a fixed value. It is deterministic.

Let's say we want to define a variable for the current weather temperature.

It has a fixed value. It is deterministic.

Let's say we want to define a variable for the outcome of rolling a dice.

It is not fixed and may have any value from 1 to 6. It is Probabilistic.

An Ordinary Variable or Deterministic Variable represents fixed information.

A Random Variable represents uncertain outcomes in the future.

EXAMPLES:

- If the outcome is fixed and known beforehand is ordinary Variable
 - If the outcome is unknown and depends on chance is Random Variable
 - The speed of your car at a given time is Ordinary Variable
 - The speed of the next car passing by on the highway is Random Variable
-

DEFINITION

A **random variable** is a quantity whose value is determined by the outcomes of the random experiments.

We can define a random variables as a function mapping each outcome in a probability space to a real number.

Given a probability space $(\Omega, \mathcal{F}, P)$, a **random variable** X is a function from the sample space Ω to the real numbers $\mathbb{R}$.

Once the outcome $\omega \in \Omega$ of the experiment is revealed, the corresponding $X(\omega)$ is known as a realization of the random variable.

DEFINITION

A **random variable** is also called a **chance variable**, a **stochastic variable** or a **variate**.

Random variables are usually denoted by capital letters such as X, Y, Z.

TYPERS OF RANDOM VARIABLES:

There are two types of variables:

1. Discrete Random Variable
2. Continuous Random Variable

TYPERS OF RANDOM VARIABLES:

There are two types of variables:

1. Discrete Random Variable

- a) Number of defective bulbs in a box of 100 bulbs,
- b) Number of calls received at a helpline center in one hour,
- c) Number of cars crossing a toll plaza in 10 minutes

2. Continuous Random Variable

- a) Example: Daily Temperature in a City,
 - b) Time taken by a student to solve a math problem,
 - c) Amount of rainfall in a day
-

EXAMPLE:

Suppose that our experiment consists of tossing 3 fair coins.

If we let Y denote the number of heads that appear,

then Y is a random variable taking on one of the values 0, 1, 2 and 3 then,

$$Y(T, T, T) = 0, Y(H, T, T) = 1$$

The outcomes related to $Y = 1$ are $(H, T, T), (T, H, T), (T, T, H)$.

EXAMPLE:

The sample space giving a detailed description of each possible outcome when three electronic components are tested may be written

$$S = \{NNN, NND, NDN, DNN, NDD, DND, DDN, DDD\},$$

where N denotes non-defective and D denotes defective.

One is naturally concerned with the number of defectives that occur. Thus, each point in the sample space will be assigned a numerical value of 0, 1, 2, or 3.

These values are random quantities determined by the outcome of the experiment.

Hence, the random variable X represents the number of defective items when three electronic components are tested.

NOTATIONS:

We shall use a capital letter, say X , to denote a random variable and its corresponding small letter, x in this case, for one of its values.

In the electronic component testing, we notice that the random variable X assumes the value 2 for all elements in the subset

$$E = \{DDN, DND, NDD\}$$

of the sample space S .

That is, each possible value of X represents an event that is a subset of the sample space for the given experiment.

EXAMPLE:

Suppose that our experiment consists of tossing 3 fair coins. If we let Y denote the number of heads that appear, then Y is a random variable taking on one of the values 0, 1, 2 and 3 with probability

$$P(Y = 0) = P\{(T, T, T)\} = \frac{1}{8}$$

$$P(Y = 1) = P\{(H, T, T), (T, H, T), (T, T, H)\} = \frac{3}{8}$$

$$P(Y = 2) = P\{(H, H, T), (H, T, H), (T, H, H)\} = \frac{3}{8}$$

$$P(Y = 3) = P\{(H, H, H)\} = \frac{1}{8}$$

Since Y must take on one of the values 0 through 3, we must have

$$1 = P\left(\bigcup_{i=0}^3 [Y = i]\right) = \sum_{i=0}^3 P(Y = i)$$

which, of course, is in accord with the preceding probabilities.

DISCRETE PROBABILITY DISTRIBUTION:

The set of ordered pairs $(x, f(x))$ is a **probability function**, **probability mass function**, or **discrete probability distribution** of the discrete random variable X if, for each possible outcome x ,

1. $f(x) = P(X = x) = P\{\zeta \in \Omega | X(\zeta) = x\}.$
2. $f(x) \geq 0,$
3. $\sum_x f(x) = 1,$

EXAMPLE:

A shipment of 20 similar laptop computers to a retail outlet contains 3 that are defective. If a school makes a random purchase of 2 of these computers, find the probability distribution for the number of defectives.

Let X be a random variable whose values x are the possible numbers of defective computers purchased by the school. Then x can only take the numbers 0, 1, and 2.

Now

$$f(0) = P(X = 0) = \frac{\binom{3}{0}\binom{17}{2}}{\binom{20}{2}} = \frac{68}{95},$$

$$f(1) = P(X = 1) = \frac{\binom{3}{1}\binom{17}{1}}{\binom{20}{2}} = \frac{51}{190}$$

$$f(2) = P(X = 2) = \frac{\binom{3}{2}\binom{17}{0}}{\binom{20}{2}} = \frac{3}{190}$$

Thus, the probability distribution of X is

x		0	1	2
$f(x)$		$\frac{68}{95}$	$\frac{51}{190}$	$\frac{3}{190}$

EXAMPLE (REVISITED):

Suppose that our experiment consists of tossing 3 fair coins. If we let Y denote the number of heads that appear, then Y is a random variable taking on one of the values 0, 1, 2 and 3 with probability

$$P(Y = 0) = P\{(T, T, T)\} = \frac{1}{8}$$

$$P(Y = 1) = P\{(H, T, T), (T, H, T), (T, T, H)\} = \frac{3}{8}$$

$$P(Y = 2) = P\{(H, H, T), (H, T, H), (T, H, H)\} = \frac{3}{8}$$

$$P(Y = 3) = P\{(H, H, H)\} = \frac{1}{8}$$

Find $P(Y \geq 2)$, $P(Y < 2)$, $P(2 \leq Y \leq 3)$.
